

DIGITAL TWIN FOR WELL-TO-SURFACE OPTIMIZATION

Cyclic Steam Stimulation (CSS) & Sucker Rod Pump (SRP) Integrated Co-Optimization for Baghewala Field

Complete System Architecture, Mathematical Methodology, Technology Stack, and Detailed Feature & Button Reference Manual.

Field Location:	Baghewala Field, Rajasthan, India
Producing Reservoir:	Jodhpur Sandstone Formation
Crude Characterization:	17.5° API Heavy Crude (3,200 cP @ 47°C, 14.5% Asphaltene)
Software Platform:	React 18 + TypeScript + Vite + HTML5 Canvas Engine
Document Version:	v1.0 Final Technical & Operational Manual
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1. Executive Summary & Field Background

1.1 Reservoir Context (Baghewala Field, Rajasthan)

The Baghewala Field in Rajasthan produces heavy crude oil (17.5° API) from the Jodhpur Sandstone formation at a setting depth of approximately 1,100 meters. The reservoir presents extreme operational challenges:

- **Ultra-High Crude Viscosity:** 3,200 cP at natural reservoir temperature (47°C).
- **High Asphaltene Content:** 14.5% asphaltene, causing precipitation risks and high viscous friction drag.
- **Low Primary Energy:** Low reservoir temperature (46–48°C) and low reservoir pressure (35 bar).
- **Poor Primary Mobility:** Requires thermal Enhanced Oil Recovery (EOR) via Cyclic Steam Stimulation (CSS) combined with Sucker Rod Pumping (SRP) artificial lift.

1.2 The Operational Problem: Unintegrated Optimization

Currently, CSS thermal injection cycles and SRP operational pumping speeds are optimized separately based on historical experience. As the reservoir cools following steam injection:

1. Crude viscosity increases exponentially back toward baseline (3,200 cP).
2. Viscous drag on the downstroke increases dramatically, overriding natural rod string gravity ($\text{Viscous Drag} > \text{Net Buoyant Weight}$).
3. This results in **Rod Floating**, polished rod slack, rod string buckling, tubing wear, fluid pound, high Steam-Oil Ratio (SOR), excessive electrical power consumption, and frequent pump failures.

1.3 The Solution: Integrated Digital Twin

This software platform delivers an industrial physics-informed AI Digital Twin that continuously models and co-optimizes both CSS thermal injection cycles and SRP artificial lift speed profiles in real time, eliminating rod floating, reducing SOR, and maximizing net economic margin (\$ NPV).

2. Software Frameworks, Tools & Libraries Used

Layer / Component	Technology / Tool	Version	Implementation Role & Description
Frontend Framework	React	v18.3.1	Component-based UI framework for state management and interactive views.
Programming Language	TypeScript	v5.5.3	Strictly typed codebase preventing runtime null checks and type mismatches.
Build Tool & Dev Server	Vite	v8.2.2	Lightning-fast bundler delivering sub-second HMR and production builds.
Graphics & Physics Engine	HTML5 2D Canvas API	Standard	Sub-pixel 60 FPS animation engine rendering beam kinematics, downhole valves, and thermal heat plume.
Design System	Vanilla CSS3 + Glassmorphism	CSS3	Custom dark mode design tokens with background blur effects and neon indicators.
Icon Suite	Lucide-React	v0.344.0	High-clarity vector icons for navigation, indicators, and alerts.
Typography	Google Fonts (Outfit & JetBrains Mono)	Web Fonts	Clean sans-serif headings and monospace numerical telemetry formatting.
Data Engine	CSV & HTML5 Blob API	Standard	In-app CSV dataset previewer, data parser, and instant 1-click dataset exporter.

3. Mathematical Methodology & Physical Equations

3.1 Andrade Temperature-Viscosity Correlation

$$\mu(T) = \mu_{ref} \cdot \exp[B \cdot (1 / (T + 273.15) - 1 / (T_{ref} + 273.15))]$$

Where:

- μ_{ref} = 3,200 cP at T_{ref} = 47°C (Ambient Baghewala Reservoir)
- B = 4,650 K (Activation energy coefficient for Jodhpur crude)
- At peak steam temperature (295°C), viscosity drops to 18 cP.

3.2 Thermal Reservoir Heat Decay Model T(t)

$$T(t) = T_{\text{res}} + (T_{\text{peak}} - T_{\text{res}}) \cdot \exp[- (\lambda / \sqrt{(V_{\text{steam}} / 2000)}) \cdot t]$$

Where:

- $T_{\text{res}} = 47^{\circ}\text{C}$, $T_{\text{peak}} = 295^{\circ}\text{C} - (\text{Soak Days} \times 3.5)$
- $\lambda = 0.045 \text{ day}^{-1}$ (Thermal conductivity decay constant)

3.3 SRP Polished Rod Loads & Rod Floating Criterion

$$W_{\text{rod,net}} = W_{\text{rod,air}} \cdot (1 - \rho_{\text{fluid}} / \rho_{\text{steel}})$$

$$\text{Upstroke Load: } F_{\text{up}} = W_{\text{rod,net}} + F_{\text{fluid}} + F_{\text{accel}} + F_{\text{viscous,up}}$$

$$\text{Downstroke Load: } F_{\text{down}} = W_{\text{rod,net}} - F_{\text{accel}} - F_{\text{viscous,down}}$$

Rod Floating Condition:

When Viscous Drag $F_{\text{viscous,down}} \propto \mu(T) \cdot v_{\text{down}}$ exceeds net buoyant gravity $W_{\text{rod,net}}$, downstroke tension drops to 0 lbs, causing rod floating, slack line, and severe buckling.

4. Detailed Feature & Button Reference Manual

4.1 Global Navigation Header & Simulation Bar

- **3D Digital Twin** : Switches to the interactive wellbore schematic and animated SRP walking beam view.
- **Dynamometer & SRP** : Switches to real-time Load vs Position Dynamometer card plots and AI anomaly classification.
- **CSS Thermal Optimizer** : Switches to steam volume injection controls, soak period tuning, and thermal decay curves.
- **What-If Lab** : Switches to parametric scenario simulation comparing Historical Practice vs Digital Twin Strategy.
- **Economic ROI** : Switches to financial daily margins, OPEX breakdown, and executive report downloader.
- **Data Center** : Switches to empirical dataset previews and 1-click CSV file downloaders.
- **LIVE TICK / PAUSE** : Toggles automatic 2.5-second time step advancement across the 60-day cycle trajectory.
- **Reset (RotateCcw)** : Instantly resets simulation day back to Day 1.
- **Auto-Mitigate (Set 3.8 SPM)** : Appears in the critical alert banner during rod floating to immediately reduce SPM speed to a safe operating rate.

4.2 Module 1: 3D Digital Twin Visualizer

Renders sub-pixel animated surface pumping unit kinematics, sucker rod string motion, polished rod stress, downhole valve action (traveling & standing valves), and reservoir steam chamber plume expansion/cooling. Includes HUD floating badges for Bottom Hole Temp (°C), Crude Viscosity (cP), Peak/Min Loads (lbs), and SPM.

4.3 Module 2: Dynamometer Card & AI Classifier

Displays Surface Load (lbs) vs Position (inches) alongside Downhole Pump Card. Features test pattern selector buttons (**AUTO** , **NORMAL** , **ROD_FLOATING** , **FLUID_POUND**) and AI neural network anomaly classification with confidence scoring.

4.4 Module 3: CSS Thermal Optimizer

Provides sliders for Steam Volume (1,000–4,500 m³), Soak Period (1–12 Days), and Production Cut-off Threshold (2.0–8.0 m³/d). Features an **Apply AI Recommendation** button to apply Pareto-optimal parameters.

4.5 Module 4: SRP Speed Tuner & Reliability Monitor

Features an **Enable Auto-Tuner** toggle button for dynamic viscosity-driven speed regulation, SPM slider, stroke length slider, plunger size slider, downstroke tension margin meter, and cycle day SPM schedule table.

4.6 Module 5: What-If Scenario Sandbox

Interactive laboratory comparing Scenario A (Historical Practice) vs Scenario B (Digital Twin Strategy) with live side-by-side performance matrix for Oil Yield, SOR, Fuel Expenses, and Net Margin.

4.7 Module 6: Economic ROI Dashboard

Tracks Net Daily Margin (\$/day), SOR ($\text{\$/m}^3/\text{\$/m}^3$), Specific Energy Cost (\$/bbl), and features an

[Export Executive Report \(JSON\)](#) button for generating official executive reports.

4.8 Module 7: Data Center & Datasets

Houses all four empirical datasets used in the digital twin:

1. [Baghewala_Reservoir_Core_Data.csv](#)
2. [CSS_Thermal_Cycle_Telemetry.csv](#)
3. [SRP_Dynamometer_Card_Time_Series.csv](#)
4. [Andrade_Viscosity_Temperature_Curve.csv](#)

Includes individual [Download CSV](#) buttons and a [Download All Datasets \(CSV Bundle\)](#) button.